

Effect of aqueous extracts of *Enicostema axillare* and *Toddalia asiatica* on complement system alternative pathway activity

Balaji Pandiyan^{1*}, Saravanan Nadanam², Venkatachalam Venkatachalam¹, Sivakkumar Thillainayagam¹ & Kannan Kamarajan¹

¹Department of Pharmacy, Faculty of Engineering & Technology, Annamalai University, Annamalainagar-608 002, Tamil Nadu, India

²Division of Biochemistry, Rani Meyyammai College of Nursing, Annamalai University, Annamalainagar-608 002, Tamil Nadu, India

Received 03 August 2015; revised 23 November 2015

Complement system, precisely, the Alternative complement pathway (AP) is the first line of defence against microbial infection. Herbal extracts are known to suppress as well as boost these pathways and have applications accordingly. Here, we explored the effect of aqueous extracts of two commonly used medicinal plants *Enicostema axillare* (Poir. ex Lam.) A.Raynal (AEEA) and *Toddalia asiatica* (L.) Lam. (AETA), on *in vitro* and *in vivo* AP activity and also compared factor B activity with that of the known immunostimulant levamisole. The *in vitro* results revealed a concentration dependent increase in the AP for levamisole, AEEA and AETA. The *in vivo* AP and factor B were studied using rat animal model. The studies were carried out independently using Veronal Buffer Saline (VBS)-AP buffer and Triethanolamine (TEA)-AP buffer to compare the suitability of TEA buffer. The AP activity in terms of concentration of the sample required to cause 50 % hemolysis (APC50) and factor B activity of AEEA and AETA treated animals were significantly ($P < 0.05$) higher than control. However, AP 50 was significantly ($P < 0.05$) higher for levamisole when compared with that of AEEA or AETA. Significant increase in Factor B activity was observed for AETA, AEEA and levamisole when compared with that of control. Factor B activity of AETA was comparable with that of levamisole ($P < 0.05$) where as factor B activity of AEEA treated animal was significantly less than levamisole ($P < 0.05$). The results suggest VBS buffer to be more sensitive than TEA buffer.

Keywords: Alternative pathway, Complement system, Factor B, Haemolytic activity

Microbial infection is a major health hazard, particularly in developing countries. Living beings protect themselves from infectious diseases by various defence mechanisms collectively known as immune system. Complement system (CS) is a part of immune system, consisting nine different proteins denoted by C1-C9 which are present in human beings and animals in inactive form¹⁻⁴. These proteins can be activated by three routes: classical pathway (CP), alternative pathway (AP) and lectin pathway (LP). The alternative pathway of the complement system is innate immune system, non-specific and considered as the first line defence against microbial infection⁵⁻⁷. Excessive complement activation is observed in inflammatory condition as well as in autoimmune disease⁸⁻¹⁰.

Many synthetic agents and herbal extracts have been reported to activate the complement system CP¹¹⁻¹³. There are only few reports available on agents that enhance AP activity^{14,15}. Many plants are reported to decrease alternate pathway activity, thereby finding

application as immunosuppressant in organ transplantation^{16,17}. There are many reports on use of herbal drugs to increase the AP activity in fish and in fish hatchery to prevent infection^{18,19}.

In ayurvedic system, the rasayanas are used to strengthen the body defence mechanism and many plants have been recommended in alternate system of medicine for the treatment of fever, cough, cold, etc. Herbal preparations exhibiting protective effect against infection may not necessarily have bactericidal/bacteriostatic effect as has been demonstrated earlier using *Caenorhabditis elegans* infection model²⁰. Hence, the herbal drugs used in the treatment of infection may, in fact, strengthen or increase the first line defence mechanism, the alternative complement pathway (AP).

In this study, we evaluated the effect of two medicinal plants, the Indian white head *Enicostema axillare* and the Orange climber *Toddalia asiatica*, commonly used in traditional medicine, on alternative complement pathway activity in search of a potential herbal extract to improve body defence mechanism against infection.

*Correspondence:

E-mail: pharmacology_balaji@yahoo.co.in

Materials and Methods

Leaves of *Enicostema axillare* (Poir. ex Lam.) A. Raynal and root bark of *Toddalia asiatica* (L.) Lam. were collected from Tirunelveli district, Tamilnadu, India and authenticated by the CCRAS-Clinical Research Unit (Siddha), Palayamkottai, Tamilnadu, India. Voucher specimens of the plants have been deposited in the Department of Pharmacy, Annamalai University.

Levamisole, the standard drug was a gift sample from MMC healthcare Limited, Chennai. Other chemicals/drugs and their respective sources were as follows: Cyclophosphamide (Endoxan Injection), German Remedies (Mumbai, India); Triethanolamine (TEA), Qualigens Fine Chemicals, India; and EGTA [ethyleneglycol-bis-(2-aminoethyl ether) N,N,N',N'-tetraacetic acid], Sigma Aldrich, USA. Veronal (VBS-AP) buffer with Mg^{++} and EGTA containing 10 mM barbital, 145 mM NaCl, 0.5 mM $MgCl_2$ and 5 mM EGTA, pH 7.4 ± 0.2] was from Boston Bioproducts (Ashland, USA). Anticoagulant, sodium citrate 3.8% w/v solution and Phosphate Buffer saline (PBS, pH 7.2) was from Himedia Laboratories Pvt. Ltd. India. Normal saline was from Baxter (India) Pvt. Ltd, India. All other chemicals used were of GR/AR grade. Human serum (HS) samples were obtained from healthy volunteers.

Experimental animals

Healthy male Wistar rats weighing between 150-200 g (from animal house, Rajah Muthiah Medical College, Annamalai University, Tamilnadu. Reg. No.160/1999/CPCSEA – Proposal No:1026), were kept in polyacrylic cages (@ 6/cage) and maintained under standard housing conditions of temperature (24-27°C) and humidity (60-65%) with 12:12 light:dark cycles. They were fed with standard pellet diet (D.S. Trading Company, Mumbai, India) and water *ad libitum*. The animals were acclimatized to laboratory conditions for 48 h prior to the experiment to minimize non specific stress.

Preparation of extract

Aqueous extracts of dried leaves of *E. axillare* and dried root bark of *T. asiatica* were prepared as described earlier²¹. Briefly, repeated extraction of the powdered plant material with boiling distilled water (1:20, w/v) till colourless extracts was obtained. The combined extract was concentrated in a rotary evaporator and Lyophilised. The yield was 28.48% w/w for aqueous extract of *E. axillare* (AEEA), 19.04% w/w for aqueous extract of *T. asiatica* (AETA).

Preparation of 1% Rabbit erythrocytes (RbE)

Fresh Rabbit blood was collected in a sterile bottle with anticoagulant (blood: anticoagulant, 9:1 v/v), mixed and centrifuged at 2000 rpm for 10 min. Supernatant was removed by decantation, cells were washed twice with PBS and twice with VBS-AP buffer (VBS/ Mg_{2+} /EGTA). About 1% v/v suspension of RbE in VBS-AP buffer was prepared by suspending washed erythrocytes in VBS-AP buffer.

Similarly, RbE suspension in TEA-AP buffer was prepared by suspending washed erythrocytes in TEA-AP buffer containing 20 mM TEA, 0.15 M sodium chloride (NaCl), 0.8 mM Sodium azide, 8 mM ethyleneglycol-bis-(β -aminoethyl ether) N,N,N',N'-tetraacetic acid and 2 mM magnesium chloride²².

In vitro alternate pathway haemolytic activity

The assay was performed in flat-bottom 96-well microtitre plates (Tarsons – 941196, Kolkata, India). 5% w/v solutions of Levamisole (Standard drug), AEEA and AETA, in triplicate, were prepared separately in VBS-AP buffer. Further dilutions were made in the micro-centrifuge tubes (1:4, 1:8, 1:16, 1:32, 1:64, 1:128, 1:256) with the VBS-AP buffer resulting in a final volume of 100 μ L in each tube and 25 μ L of HS was added to each tube. After incubating for 30 min at 37°C, 25 μ L RbE suspension was added to each tube and the tubes were incubated at 37°C for 60 min. Subsequently, the tubes were centrifuged at 1000rpm for 6 min. About 50 μ L of the supernatant was transferred to a flat-bottom microtiter plate (Tarsons – 941196, Kolkata, India), mixed with 200 μ L water and the absorbance was measured at 412 nm in an ELISA automatic plate reader (Multiscan EX, Thermo Fisher scientific, India). Controls in this assay consisted of RBC incubated in distilled water (Total lysis), RBC incubated in buffer (Blank) and the colour of HS-dilution (complement blank). The absorbance of complement blank was subtracted from absorbance values of test sample to get the corrected absorbance of test serum^{16,18,22-25}.

Percentage haemolysis for each dilution was calculated by the following formula:

$$\text{Percentage haemolysis (y)} = \frac{(\text{Corrected absorbance of test sample} - \text{Absorbance of blank})}{(\text{Absorbance of Total lysis} - \text{Absorbance of blank})} \times 100$$

The percentage hemolysis was plotted against concentration of the drug/extract. The concentration of drug/extract required for producing 50% haemolysis (APC50) was calculated from the graph.

A similar study was carried out using TEA-AP buffer in place of VBS-AP buffer (APC50).

***In vivo* complement system AP50**

Grouping of animals

Wistar male rats were randomized into four groups of six animals each and treated with drug/extract for 14 days as given in Table 1.

At the end of 14 days, blood samples were collected from all animals by puncturing the retro orbital plexus. Serum was separated from blood by centrifugation and 1:25 dilution was made in VBS-AP buffer. Series of dilutions, in VBS-AP buffer, of each 1:25 diluted serum sample (1:10; 1:5; 1:4; 1:3; 1:2; 1:1; 2:1; 3:1; 4:1; 5:1; 10:1; 10:0) was made resulting in final volume of 50 μ L in each tube. 50 μ L RbE suspensions in VBS-AP buffer was added to each tube^{11,26}. Control tubes consisted of 50 μ L RbE for total lysis, 50 μ L RbE and 50 μ L VBS-AP buffer for blank. Incubated at 37°C for 30 min with intermittent agitation. To all serum tubes and blank tube, 150 μ L ice-cold normal saline was added and to total lysis tubes 200 μ L distilled water was added. All the tubes were centrifuged at 1000 rpm for 6 min. 200 μ L supernatants were transferred to flat-bottomed 96-well plate. Absorbance of the supernatants was measured at 412 nm in ELISA automatic plate reader (Multiscan EX, Thermo Fisher scientific, India)^{11,26-28}.

Percentage haemolysis for each dilution was calculated by using the following formula:

$$\text{Percentage haemolysis (y)} = \frac{(\text{Absorbance of test serum} - \text{Absorbance of blank})}{(\text{Absorbance of Total lysis} - \text{Absorbance of blank})} \times 100$$

Mean percentage lysis was plotted against the serum dilution. Dilution required for 50% haemolysis (APD50) was noted.

A similar study was carried out using TEA-AP buffer in place of VBS-AP buffer (APD50).

Evaluation of factor B activity

Rat serum depleted of factor B was prepared by keeping rat sera at exactly 56°C²⁸. Every 30 s, 60 μ L serum was removed and transferred to tubes containing 300 μ L VBS-AP buffer. Subsequently, 200 μ L of the rabbit erythrocyte suspension was

added, and the mixture was incubated at 37°C for 30 min. Cold PBS was added to stop the reaction, and the samples were then centrifuged for 10 min²⁹. The absorbance of the solution at 412 nm was measured³⁰, and the percent lysis was calculated in reference to the lysis of erythrocytes in water (set to 100%) Based on the time dependent haemolysis factor B assessment was done by using modified Mayer's method³¹.

Briefly, 60 μ L of RB or 60 μ L of VBS-AP was added to 300 μ L each of test serum diluted 1:5 with VBS-AP buffer and incubated with 200 μ L rabbit erythrocyte suspension for 30 min at 37°C. After incubation, 300 μ L cold PBS was added, the samples were centrifuged and 200 μ L of the supernatant was transferred to 96-well plates. The absorbance was measured at 412 nm using ELISA automatic plate reader (Multiscan EX, Thermo Fisher scientific, India). The absorbance of cell supernatants lysed in water was noted as 100% lysis and incubated with VBS-AP buffer was noted as blank (0% lysis). Factor B activity was calculated by subtracting % lysis of test serum from % lysis of test serum with RB. A similar study was carried out using TEA-AP buffer in place of VBS-AP buffer.

The data was analysed using one-way analysis of variance (ANOVA), followed by Dunnett's t – test. $P < 0.05$ was considered significant.

Results

The *in vitro* studies revealed that there is a concentration dependent increase in the AP activity for the standard drug levamisole, AEEA and AETA. The APC50 value was found to be 313.54 ± 7.47 , 435.50 ± 7.5 and 404.13 ± 1.67 μ g/mL for levamisole, AEEA and AETA, respectively with VBS-AP buffer (Fig. 1A). Similarly with TEA-AP buffer, the APC50 was 343.49 ± 2.17 , 493.46 ± 2.85 and 454.40 ± 1.93 μ g/mL for levamisole, AEEA and AETA, respectively (Fig. 1B).

In vivo study also exhibited similar results with APD50 value of 0.03 ± 0.01 , 7.52 ± 0.05 , 2.20 ± 0.08 and 3.15 ± 0.02 for control, levamisole, AEEA and AETA treated groups, respectively with VBS-AP buffer (Fig. 2A) and 0.02 ± 0.01 , 5.81 ± 0.37 , 1.93 ± 0.18 and 3.04 ± 0.05 for control, levamisole, AEEA and AETA treated groups, respectively with TEA-AP buffer (Fig. 2B).

The percentage lysis was plotted against heating time. Baseline hemolytic activity was obtained when heated up to 3.3 min (210 s). Hence, sera heated at 56°C for 3.3 min was considered as rat serum

Table 1 — Grouping and treatment of animals

Groups	Treatments
Control	Received normal saline 10 mL/kg body wt. for 14 days
LEV	Received 50 mg/kg body wt., p.o. for 14 days
AEEA	Received 200 mg/kg body wt., p.o. for 14 days
AETA	Received 100 mg/kg body wt. p.o for 14 days

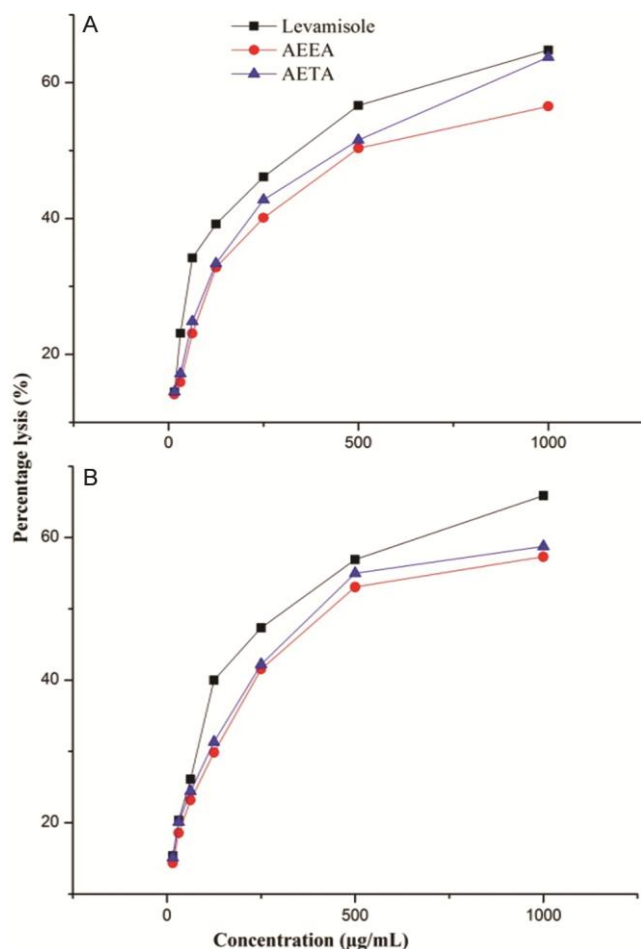


Fig. 1 — *In vitro* AP activity expressed as a percentage haemolysis using (A) VBS-AP buffer; and (B) TEA-AP buffer. [Values are expressed as mean $\pm$ S.E.M, n=6 in each group, ^a $P < 0.01$ AEEA and AETA when compared with Levamisole]

depleted factor B (RB) and used for further studies to evaluate factor B activity (Fig. 3).

The factor B enhancement activity of AETA is comparable with that of levamisole ($P < 0.05$). AEEA factor B enhancement activity is significantly less when compared with levamisole ($P < 0.05$), although it is having significant activity when compared with control ($P < 0.05$) indicating AEEA also has AP enhancement activity and factor B enhancement activity though less than that of levamisole (Table 2).

Discussion

Our previous finding revealed that AEEA and AETA have immunostimulant activity²¹ and the work was extended to evaluate their effect on AP activity in terms of haemolytic activity as well as factor B activity, since it is the first line defence against infections. Both *in vivo*¹¹ and *in vitro*^{15,16,26}

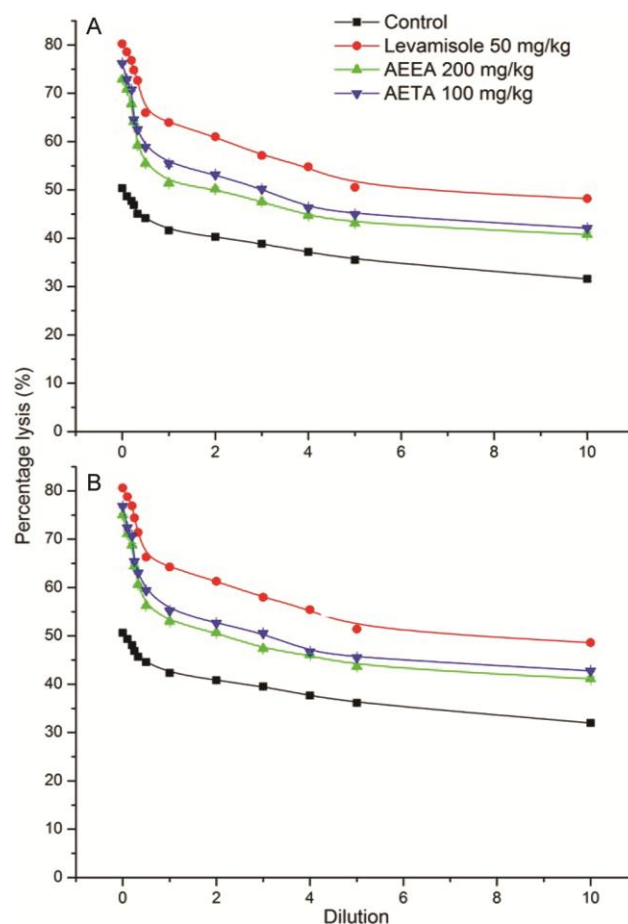


Fig. 2 — *In vivo* AP activity expressed as a percentage haemolysis using (A) VBS-AP buffer; and (B) TEA-AP buffer. [Values are expressed as mean $\pm$ S.E.M, n=6 in each group, ^a $P < 0.01$ Levamisole, AEEA 200 mg/kg and AETA 100 mg/kg when compared with control. X axis denotes number of mL of TEA-AP buffer added for each mL of diluted serum]

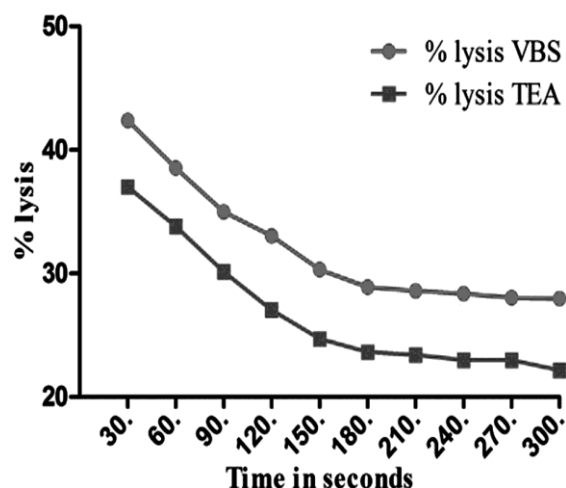


Fig. 3 — Effect of heating time on Factor B activity

Table 2 — Factor B activity

Groups	Percentage lysis (%)					
	Without factor B depleted serum		With factor B depleted serum		Mean difference in factor B activity	
	(A)		(B)		(B – A)	
	VBS	TEA	VBS	TEA	VBS	TEA
Control	33.55±0.29	27.50±0.20	36.04±0.29	30.75±0.30	2.48±0.38	3.25±0.42
LEV	78.65±0.45	76.55±0.28	97.73±0.16	94.68±0.25	19.07±0.56*	18.13±0.32*
AEEA 200 mg/kg	68.36±0.22	67.66±0.21	83.40±0.45	82.22±0.14	15.04±0.48*	14.55±0.31*
AETA 100 mg/kg	73.60±0.27	70.18±0.19	91.51±0.19	87.10±0.22	17.91±0.43*	16.91±0.11*

Values are expressed as mean± S.E.M, n=6 in each group. ^a $P < 0.05$ Levamisole, AEEA 200 mg/kg and AETA 100 mg/kg when compared with control.

experimental models have been used to evaluate AP activity. VBS-AP buffer is commonly used for evaluation of complement activity and there are few reports on the use of TEA-AP buffer. However, after inclusion under banned drugs category, procuring VBS-AP buffer has become a difficult task and also became more expensive. Hence, we explored the utility of TEA-AP buffer as a better alternative of VBS-AP buffer.

In agreement with earlier findings³²⁻³⁵, the present study also revealed that levamisole increased the AP activity significantly ($P < 0.05$) when compared to control. Similarly, AEEA and AETA also significantly increased the AP activity but their effects were significantly less than that of levamisole. Both *in vitro* and *in vivo* studies exhibited similar effect.

Interestingly, *in vitro* and *in vivo* AP enhancement activity of AETA was significantly less than that of levamisole ($P < 0.05$) but factor B enhancement activity of AETA was comparable with that of levamisole ($P < 0.05$). This may be due to the fact that AETA acts specifically only on factor B, whereas levamisole, a well known immunostimulant, may act on many other sites apart from factor B. AETA potentially activating only factor B which leads to cleavage of factor B to Ba and Bb. Bb is involved in the proliferation of pre-activated B lymphocytes, while Ba inhibits their proliferation thereby causing homeostasis of B lymphocyte activity. It is possible that uncontrolled stimulation of AP by levamisole may cause unwanted side effects like inflammatory reactions and autoimmune type of reactions. Thus, AETA might be a better agent to defend against infections without producing inflammatory disorders and autoimmune type of reactions when compared with levamisole.

AEEA factor B enhancement activity was significantly less when compared with levamisole ($P < 0.05$), although it was having significant activity when compared with control ($P < 0.05$) indicating AEEA also has AP enhancement activity

and factor B enhancement activity though less than that of levamisole.

The results also revealed that the haemolysis was significantly higher ($P < 0.05$) when VBS-AP buffer was used than with TEA-AP buffer although the final result of AP activity was similar.

From the present study, it can be inferred that the simple *in vitro* procedure using cost effective TEA-AP buffer can be used for preliminary screening of large number of samples and further confirmation can be done by *in vivo* studies using sensitive but costlier VBS-AP buffer.

In accordance with widely believed concept that whole plant extract gives a balanced biological effect than a pure synthetic or pure drug from plant, the present study revealed that the root bark extract of *Toddalia asiatica* (AETA) increased Alternative complement pathway (AP) activity compared with the common drug levamisole.

Acknowledgement

The authors express their gratitude to University Grants Commission, New Delhi, India for their financial support to this work under Major Research Project Scheme.

References

- Walport MJ, Complement. First of two parts. *N Engl J Med*, 344 (2001) 1058.
- Anne K, Mario H, Eliska S & Mihaly J, Factor H: A complement regulator in health and disease, and a mediator of cellular interactions. *Biomolecules*, 2 (2012) 46.
- Zewde N, Gorham RD Jr, Dorado A & Morikis D, Quantitative modeling of the alternative pathway of the complement system. *PLoS ONE*, 11 (2016) e0152337.
- Daniel Ricklin, Edimara S Reis & John D Lambris, Complement in disease: a defence system turning offensive. *Nat Rev Nephrol*, 12 (2016) 383.
- Morten Harboe, Ebbe Billmann Thorgersen & Tom Eirik Mollnes, Advances in assay of complement function and activation. *Adv Drug Delivery Rev*, 63 (2011) 976.
- Wen-Chao Song, Maria Rosa Sarrias & John DL, Complement and innate immunity. *Immunopharmacol*, 49 (2000) 187.

- 7 Jason R Dunkelberger & Wen-Chao Song, Complement and its role in innate and adaptive immune responses. *Cell Res*, 20 (2010) 34.
- 8 Dorota Tichaczek-Goska, Deficiencies and excessive human complement system activation in disorders of multifarious etiology. *Adv Clin Exp Med*, 21 (2012) 105.
- 9 Maria VC & Robert BS, Complement in health and disease. *Adv Drug Delivery Rev*, 63 (2011) 965.
- 10 Nibhriti Das, Complement and membrane-bound complement regulatory proteins as biomarkers and therapeutic targets for autoimmune inflammatory disorders, RA and SLE. *Indian J Exp Biol*, 53 (2015) 701.
- 11 Ana Paula Landi Librandi, Tais Nader Chryso stomo, Ana Elisa CSA, Carem Gledes Vargas Recchia, Sergio Akira Uyemura & Ana Isabel de Assis-Pandochi, Effect of the extract of the tamarind (*Tamarindus indica*) fruit on the complement system: Studies *in-vitro* and in hamsters submitted to a cholesterol-enriched diet. *Food Chem Toxicol*, 45 (2007) 1487.
- 12 Batbayar N, Banzragch D, Inngjerdigen KT, Naran R, Michaelsen TE & Paulsen BS, Polysaccharides from Mongolian plants and their effect on the complement system: I. Polysaccharides from plants of the *Asteraceae* family. *Asian J Tradit Med*, 3 (2008) 33.
- 13 Perricone R, Carolis CDE, Moretti C, Santuarit E, Sanctis GDE & Fontana L, Complement, complement activation and anaphylatoxins in human ovarian follicular fluid. *Clin exp Immunol*, 82 (1990) 359.
- 14 Chen X & Ma B-L, *Trichosanthin*, an initiator of the alternative complement activation pathway. *Clin Exp Immunol*, 93 (1993) 248.
- 15 Shimizu S, Kitada H, Yokota H, Yamakawa J, Murayama T, Sugiyama K, Izumi H & Yamaguchi N, Activation of the alternative complement pathway by *Agaricus blazei* Murill. *Phytomed*, 9 (2002) 536.
- 16 Cos P, Hermans N, Van Poel B, De Bruyne T, Apers S, Sindambiwe JB, Vanden Berghe D, Pieters L & Vlietinck AJ, Complement modulating activity of Rwandan medicinal plants. *Phytomed*, 9 (2002) 56.
- 17 Kulkarni AP, Kellaway LA & Kotwal GJ, Herbal Complement inhibitors in the treatment of neuroinflammation. *Ann NY Acad Sci*, 1056 (2005) 413.
- 18 Biller-Takahashi, Jaqueline Dalbello; Takahashi, Leonardo Susumu; Marzocchi-Machado, Cleni Mara; Zanuzzo, Fabio Sabbadin; Sabioni, Rafael Estevan; Urbinati & Elisabeth Criscuolo, Hemolytic activity of alternative complement pathway as an indicator of innate immunity in pacu (*Piaractus mesopotamicus*). *R. Bras. Zootec*, 41 (2012) 237.
- 19 Amir Tukmechi, Ahmad Morshedi & Nowruz Delirezh, Changes in intestinal microflora and humoral immune response following probiotic administration in rainbow trout (*Oncorhynchus mykiss*). *J Anim Vet Adv*, 6 (2007) 1183.
- 20 Cin Kong, Wageeh A Yehye, Noorsaadah Abd Rahman, Man Wah Tan & Sheila Nathan, Discovery of potential anti-infectives against *Staphylococcus aureus* using a *Caenorhabditis elegans* infection model. *BMC Complement Altern Med*, 14 (2014) 1.
- 21 Balaji P, Venkatachalam V, Sivakkumar T & Kannan K, Immunomodulatory activity of aqueous extracts of *Enicostema Axillare* leaves and *Toddalia Asiatica* root bark. *Int J Pharma Bio. Sci*, 6 (2015) 251.
- 22 Klerx JPAM, Beukelman, CJ, Van Dijk H & Willers JMN, Microassay for colorimetric estimation of complement activity in guinea pig, human and mouse serum. *J Immunol Methods*, 63 (1983) 215.
- 23 Mqadmi A, Zheng X, Song J, Abramowitz S, Giclas P & Yazdanbakhsh K, Prevention of complement-mediated immune hemolysis by a small molecule compound. *Biochem Biophysical Res Commun*, 325 (2004) 1465.
- 24 Deharo E, Baelmans R, Gimenez A, Quenevo C & Bourdy G, *In vitro* immunomodulatory activity of plants used by the Tacana ethnic group in Bolivia. *Phytomed*, 11 (2004) 516.
- 25 Balaji P, Fathima N, Lakshyajit Kakati, Jaseer K.K, Uma Krithika S & Kannan K, Activation of complement system alternate pathway by *Andrographis paniculata*, *Ocimum sanctum* and *Azadirachta indica*. *Eur J Pharm Med Res*, 2 (2015) 401.
- 26 Smit HF, Kroes BH, Van Den Berg AJ, Van Der Wal D, Van Den Worm E, Beukelman CJ, Van Dijk H & Labadie RP, Immunomodulatory and anti-inflammatory activity of *Picrorhiza scrophulariiflora*. *J Ethnopharmacol*, 73 (2000) 101.
- 27 Smith MC, Pensky J & Naff GB, Inhibition of Zymosan-Induced Alternative Complement Pathway Activation by Concaavalin A. *Infect.Immun*, 38(3), (1982) 1279-84.
- 28 Bitencourt CS, Duarte CG & Azzolini AECS, Assis-Pandochi AI, Alternative complement pathway and factor B activities in rats with altered blood levels of thyroid hormone. *Braz J Med Biol Res*, 45 (2012) 216.
- 29 Lisbeth Bjerring Jensen & Claus Koch, An assay for complement factor B in species at different levels of evolution. *Dev Comp Immunol*, 15 (1991) 173.
- 30 Angela Martin PJ, Lachmann. Lise Halbwachs & Hobart MJ, Haemolytic diffusion plate assays for factors B and D of the alternative pathway of complement activation. *Immunochem*, 13 (1976) 317.
- 31 Mayer MM, Complement and complement fixation. In: Experimental Immunochemistry. 2nd edn. (Ed. Kabat EA & Mayer MM, Springfield: Charles C. Thomas, Publisher), 1971, 133.
- 32 Ortiz-Ortiz L, Ruth Capin, Capin NR, Sepulveda B & Zamacona G, Activation of the alternative pathway of complement by *Entamoeba histolytica*. *Clin Exp Immunol*, 34 (1978) 10.
- 33 Sajid Maqsood MH, Samoon & Prabjeet Singh, Immunomodulatory and growth promoting effect of dietary levamisole in *Cyprinus carpio* fingerlings against the challenge of *Aeromonas hydrophila*. *Turk J Fish Aquat Sci*, 9 (2009) 111.
- 34 Alishahi M, Ranjbar MM, Ghorbanpour M, Peyghan R, Mesbah M & Razi Jalali M, Effects of dietary Aloe vera on specific and nonspecific immunity of Common carp (*Cyprinus carpio*). *J Vet Res*, 4 (2010) 85.
- 35 Mojtaba Alishahi & Esmail Abdy, Effects of different levels of *Aloe vera* L. Extract on growth performance, hemato-immunological indices of *Cyprinus carpio* L. *Iranian J Vet Sci Technol*, 5 (2013) 33.